

4. CONTAINERS VS VMS: FLATS OR HOUSES

Both keep apps apart. They share different things.

ANALOGY: A HOUSE AND A FLAT

My own boiler.
It takes a while to heat.



Vera

Lives in



Detached house
own foundations,
own boiler

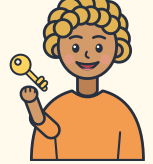
OR



Block of flats
one shared boiler,
a locked door per flat

moves in

Heating is
already on. I
just walk in.



Tom

A VM pretends to be a whole computer. A container is an ordinary process, fenced off on the host.

VM: A HOUSE WITH ITS OWN BOILER

each VM repeats these

Your app

— your program and its libraries

Guest OS

— a full OS with its own kernel, e.g. Ubuntu or Windows

Hypervisor

— fakes CPU, RAM and disks for each VM (KVM, Hyper-V, ESXi)

Hardware

— the physical server

CONTAINER: A FLAT IN ONE BUILDING

each flat has only this

Your app

— your program plus the files it needs, but no kernel

Runtime

— starts and fences the processes (containerd, runc)

Host kernel

— one Linux kernel for all: namespaces = walls, cgroups = meters

Hardware

— the physical server

THE SAME IDEA IN COMPUTERS



HOUSE = VIRTUAL MACHINE



FLAT = CONTAINER

BOILER (KERNEL)

its own, inside the guest OS

the host's Linux kernel, shared by all

START-UP

typically tens of seconds: a whole OS must boot

typically under a second: the kernel is already running

SIZE

typically gigabytes

typically tens to hundreds of megabytes

ISOLATION

stronger: a kernel bug usually stays in one VM

weaker: a kernel bug can reach every flat

OTHER OS

a Windows VM can run on a Linux host

Linux containers need a Linux kernel

TRY IT: MOVE INTO A FLAT

```
$ docker run -d -p 8080:80 nginx
```

starts nginx in the background; host port 8080 → port 80 inside. Pulls the image if missing.

```
$ docker ps
```

lists running containers: the flats someone lives in

```
$ docker images
```

lists images: the floor plans the flats are built from

FLATS INSIDE A HOUSE



macOS or
Windows



small
Linux VM



Linux
containers

Docker Desktop runs Linux containers inside a small Linux VM. In the cloud, many containers run inside VMs too.



REMEMBER: A VM brings its own kernel, like a house with its own boiler. Containers share the host kernel, like flats share one boiler.